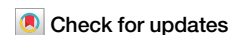


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Enhanced diffusion over a periodic trap by hydrodynamic coupling to an elastic mode



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In many physical systems, degrees of freedom are coupled *via* hydrodynamic forces, even in the absence of Hamiltonian interactions. A particularly important and widespread example concerns the transport of microscopic particles in fluids near deformable boundaries. In such a situation, the influence of elastohydrodynamic couplings on Brownian motion remains to be understood. Unfortunately, the temporal and spatial scales associated with the thermal fluctuations of usual surfaces are often so small that their deformations are difficult to monitor experimentally, together with the much slower and larger particle motion at stake. Here, we propose a minimal model describing the hydrodynamic coupling of a colloidal particle to a fluctuating elastic mode, in presence of an external periodic potential. We demonstrate that the late-time diffusion coefficient of the particle increases with the compliance of the elastic mode. Our results reveal and quantify two features: first, spontaneous microscopic transport in complex environments can be affected by soft boundaries – a situation with numerous practical implications in nanoscale and biological physics; and second, the effects of fast and tiny surface deformations are imprinted in long-term and large-distance colloidal mobility, and are therefore measurable in practice.

Brownian motion refers to the random movement of a colloid at equilibrium, induced by its collisions with the molecules of the surrounding fluid. It has played a foundational role in modern statistical physics since its discovery¹ and theoretical description^{2,3} over a century ago. The celebrated experiments of Perrin on colloids in a gravitational field⁴ validated further the theoretical description and confirmed the atomic nature of matter. Extending the classical framework of Brownian motion in bulk Newtonian fluids to more realistic systems, such as particles diffusing in complex environments, opens up new challenges for nanoscale and biological physics. One common route is to study Brownian motion in viscoelastic fluids or intracellular media, as typically performed through microrheology⁵, which revealed the emergence of anomalous diffusion⁶. Another source of complexity lies in the introduction of topological constraints and boundaries. For instance, a simple-yet-canonical situation where interfaces matter is the diffusion near rigid walls, that has been extensively studied both theoretically and experimentally^{7–15}. Besides the practical interest of such a configuration towards the sensing of surface forces¹⁶, one of the key aspects lies perhaps in the apparition of anisotropic and space-dependent mobilities¹⁷ due to the frictional boundary condition at the walls. As a consequence, multiplicative noises emerge, leading to non-Gaussian diffusion and the probability enhancement of rare events^{18–22}.

However, in most practical situations where molecules or micro-organisms diffuse near surfaces, the latter are typically soft. Therefore, the dynamical properties of Brownian particles near deformable boundaries must be understood. Along this line of thought, a few studies, mostly theoretical, have addressed the case of tracers diffusing near soft surfaces. Examples include liquid interfaces^{23,24}, and fluctuating ones²⁵. The case of elastic interfaces was also addressed in the contexts of contactless microrheology²⁶ and biological membranes²⁷. Nevertheless, these works focused on the low-coupling regime, where the particles are distant or point-like, and thus unable to generate rectified flows that lead to surface deformation. Interestingly, though, within a purely deterministic context, near-contact elastohydrodynamic couplings have been studied in the past decades. Examples include contactless colloidal-probe rheology of soft substrates in normal mode^{28,29}, as well as the emergence of lift forces at low Reynolds numbers^{30,31}, that were first revealed theoretically^{32–36}, and then confirmed experimentally^{37–41}. The leading-order corrections induced by elasticity to the full mobility tensor near a rigid wall⁷ were obtained for various geometries and rheologies^{42–44}, and revealed a zoology of counter-intuitive inertial-like solutions.

In view of the above state of the art, the arising challenge is now to understand the influence of soft boundaries on the spontaneous motion of

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nearby finite-sized Brownian particles. Molecular-dynamics simulations were performed in such a context⁴⁵. Besides, a recent experimental study indicated that soft colloids diffusing near rigid walls feel transient, non-conservative forces that could be related to an intricate combination of thermal fluctuations, soft boundaries and viscous flows⁴⁶. Finally, at a practical level, a natural question to ask is whether the effects of fast and tiny surface fluctuations can be observed over the long times and large distances associated with colloidal diffusion.

Here, we introduce a minimal, but generic, theoretical model in order to investigate Brownian elastohydrodynamic effects. Specifically, we combine the hydrodynamic coupling of a colloidal particle with an elastic mode^{47,48} to an external periodic trapping^{49–52}. We solve the problem both analytically and numerically, and demonstrate that the late-time diffusion coefficient of the particle increases notably with the compliance of the elastic mode.

Results Model

As depicted in Fig. 1, we consider an overdamped system made of two coupled degrees of freedom: i) a principal degree of freedom $q_1(t)$, corresponding to the spatial position of a test colloid over time t , which is free to explore space while being subject to a periodic trapping potential $\phi(q_1)$ with spatial period λ (in practice, such trapping potential could represent a surface with patchy, adhesive patterns); and ii) an auxiliary degree of freedom $q_2(t)$, corresponding to an elastic mode, which is harmonically confined with a potential energy $q_2^2/(2\kappa)$, where κ is the compliance. The compliance, i.e., inverse stiffness, represents the softness parameter of the elastic mode.

Thus, the total energy of the system is given by the Hamiltonian

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa}. \quad (1)$$

At thermal equilibrium, the probability density function (PDF) of q_1 and q_2 is given by the Gibbs-Boltzmann distribution

$$P_{\text{eq}}(q_1, q_2) = \frac{\exp[-\beta H(q_1, q_2)]}{Z}, \quad (2)$$

where Z is the normalizing partition function, and where $\beta = 1/(k_B T)$ with T the temperature and k_B Boltzmann's constant. The equilibrium statistics of q_1 and q_2 are independent due to the additivity of the Hamiltonian, which implies, in particular, that q_2 does not affect the equilibrium distribution of q_1 .

The system dynamics is described by the coupled overdamped Langevin equations

$$\gamma_{11} \frac{dq_1}{dt} + \gamma_{12} \frac{dq_2}{dt} = -\phi'(q_1) + \eta_1(t), \quad (3)$$

$$\gamma_{12} \frac{dq_1}{dt} + \gamma_{22} \frac{dq_2}{dt} = -\frac{q_2}{\kappa} + \eta_2(t), \quad (4)$$

where the constants γ_{ij} , with $i, j \in \{1, 2\}$, are the components of the friction tensor γ , which is related to the mobility tensor μ through $\mu = \gamma^{-1}$; and where the functions $\eta_i(t)$ are zero-mean Gaussian white noises, with a correlation function $\langle \eta_i(t) \eta_j(t') \rangle = 2k_B T \delta(t - t') \gamma_{ij}$, where $\langle \cdot \rangle$ is the ensemble average. This choice of noise correlation is self-consistent with the equilibrium distribution of q_1 and q_2 being given by Eq. (2).

Theoretical analysis

Let us now define

$$\epsilon = \frac{\kappa k_B T}{\lambda^2} \quad (5)$$

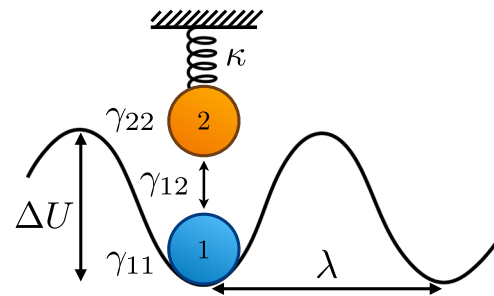


Fig. 1 | Schematic of the system. A test Brownian particle (1) is trapped in a periodic energy potential of amplitude ΔU and spatial period λ . It is coupled via a hydrodynamic-like interaction of constant damping coefficient γ_{12} to a thermally-fluctuating elastic mode, represented by another Brownian particle (2) trapped in a harmonic trap of compliance (i.e., inverse stiffness) κ . Besides, both particles feel individual hydrodynamic drag forces represented by their constant self-drag coefficients γ_{11} and γ_{22} , and are described within the overdamped Langevin regime.

the dimensionless compliance. In the weakly-compliant regime ($\epsilon \ll 1$), one can compute the late-time diffusion coefficient $D^*(\epsilon)$ of the colloid using a perturbation analysis at $\mathcal{O}(\epsilon)$, together with an exact representation of the effective diffusion coefficient obtained via the Stokes-Einstein relation, and a multiscale analysis^{53,54}. Doing so (see details in SI Note 3), we obtain

$$D^*(\epsilon) \simeq D^*(0)(1 + \alpha), \quad (6)$$

with

$$\alpha = \beta \epsilon \frac{\lambda^2 \gamma_{12}^2}{\gamma_{11}^2} \frac{\int_0^\lambda dq_1 \phi'(q_1)^2 \exp[\beta \phi(q_1)]}{\int_0^\lambda dq_1 \exp[\beta \phi(q_1)]}, \quad (7)$$

and

$$D^*(0) = \frac{k_B T \lambda^2}{\gamma_{11} Z_+ Z_-}, \quad (8)$$

where

$$Z_{\pm} = \int_0^\lambda dq_1 \exp[\pm \beta \phi(q_1)]. \quad (9)$$

Since $\alpha \geq 0$, the hydrodynamic coupling of the colloid to the elastic mode always increases the late-time diffusion coefficient of the colloid. Moreover, the effect is larger at larger compliance κ . We also see that a non-flat periodic potential $\phi(q_1)$ is required for the effect to exist, within the assumptions of the model. Such an influence of external potentials on hydrodynamic friction is reminiscent of a molecular-dynamics study in the literature⁵⁵.

In the particular case of a sinusoidal potential, where $\phi(q_1) = \Delta U \sin(2\pi q_1/\lambda)$, with ΔU the amplitude of the potential, we have

$$\alpha = 4\pi^2 \epsilon \frac{\gamma_{12}^2 v I_1(v)}{\gamma_{11}^2 I_0(v)}. \quad (10)$$

with $v = \beta \Delta U$, where I_n denotes the order- n modified Bessel function of the first kind.

As such, the relative increase of the late-time diffusion coefficient is larger for smaller λ . This is attributed to the fact that for a smaller λ the coupling to the elastic mode increases the jump rate of the colloid between neighboring minima, instead of being averaged out as would be the case at large λ . We also see that the relative increase of the late-time diffusion coefficient depends linearly on the function $v I_1(v)/I_0(v)$, which is a monotonically increasing function of v . This means that the relative magnitude of the effect increases with the depth of the trap – the obvious counterpart

being that larger times are then needed to reach the late-time diffusive regime. Finally, the relative magnitude of the effect increases with the relative magnitude γ_{12}/γ_{11} of the hydrodynamic coupling to the elastic mode, as expected.

The multiscale analysis can also be employed in the limit where $\epsilon \rightarrow \infty$, which corresponds to a very compliant elastic mode. In this case, we also find (see details in SI Note 4) that the late-time diffusion coefficient is increased relatively to the rigid-case value, and saturates to a maximum value at large ϵ , as

$$D^* = \frac{k_B T \mu_{11} (\mu_{11} \mu_{22} - \mu_{12}^2)}{\lambda^{-2} Z_+ Z_- (\mu_{11} \mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt{\epsilon}}\right). \quad (11)$$

Once again, we first note that when the periodic potential $\phi(q_1)$ is flat, the late-time diffusion coefficient is not modified by the hydrodynamic coupling to the elastic mode. However, interestingly, such a plateau value can be notably higher than the rigid-case value, indicating the potential strength of the effect in practical systems relevant to soft matter and biophysics. For instance, let us consider a spherical particle of radius $a = 100$ nm within a fluid of viscosity $\eta = 10^{-3}$ Pa.s and at temperature $T = 300$ K. The particle is assumed to be at a distance $d = 10$ nm from a soft tissue with a Young modulus $E = 1$ kPa, or equivalent stiffness $\kappa^{-1} \sim E\sqrt{ad}$, and is placed in a trapping potential of amplitude $4 k_B T$ with spatial period $\lambda = 100$ nm. Taking $\mu_{11} = \mu_{22} = (6\pi\eta a)^{-1}$ for the self-mobilities, and Oseen's $\sim \mu_{12} = (4\pi\eta r)^{-1}$ for the coupling mobility, with $r = 2a + d$, one expects a 100% relative increase of the long-term diffusivity with respect to the classical Lifson-Jackson value.

Heuristic argument

In order to understand the softness-induced late-time diffusivity enhancement from a physical point of view, we now present a much simpler heuristic argument giving the same result as the rigorous analysis in the small- ϵ regime (see details in SI Note 2). The intrinsic time scale $\tau_2 = \kappa/\mu_{22}$ of the stochastic process $q_2(t)$ is assumed to be small with respect to all the other time scales. In this limit, the effective zero-temperature overdamped Langevin equation for $q_1(t)$ is

$$\frac{dq_1}{dt} = -\mu_e(q_1)\phi'(q_1), \quad (12)$$

where we have introduced the dressed mobility

$$\mu_e(q_1) = \frac{1}{\gamma_{11} \left[1 + \frac{\kappa \gamma_{12}^2 \phi''(q_1)}{\gamma_{11}^2} \right]}. \quad (13)$$

The equilibrium marginal PDF in position q_1 is $P_{\text{meq}}(q_1) = (Z_-)^{-1} \exp[-\beta\phi(q_1)]$. Therefore, assuming that the time-dependent marginal PDF $P_1(q_1, t)$ satisfies an effective Fokker-Planck equation, the latter must have

the form

$$\frac{\partial P_1(q_1, t)}{\partial t} = \frac{\partial}{\partial q_1} \left\{ \mu_e(q_1) \left[k_B T \frac{\partial P_1(q_1, t)}{\partial q_1} + \phi'(q_1) P_1(q_1, t) \right] \right\}, \quad (14)$$

where the T -independent term on the right-hand side is determined from the zero-temperature dynamics, while the T -dependent term on the right-hand side is constructed to ensure convergence of P_1 to P_{meq} at infinite time. Note that we have implicitly assumed here that the dressed mobility $\mu_e(q_1)$ does not exhibit any temperature dependence.

The dressed diffusion coefficient is given by the Stokes-Einstein relation, $D_e(q_1) = k_B T \mu_e(q_1)$, which differs from the bare (or “microscopic”) diffusion coefficient $D_m = k_B T \mu_{11}$ of the colloid. Interestingly, the spatial dependence of the dressed diffusion coefficient implies that the effective Langevin equation must have a multiplicative noise, leading to a supplementary source of non-Gaussian displacements^{16,18–22,56} in addition to the trivial one induced by the trapping potential.

We now invoke the Lifson-Jackson formula⁵⁷, providing the late-time diffusion coefficient in a periodic potential, which reads

$$D^*(\epsilon) = \frac{\lambda^2}{\int_0^\lambda dq_1 \frac{\exp[\beta\phi(q_1)]}{D_e(q_1)} \int_0^\lambda dq_1 \exp[-\beta\phi(q_1)]}. \quad (15)$$

From this expression, we can recover Eqs. (6) and (7). We emphasize that the heuristic argument we presented here is not rigorous. Nevertheless, it can also be applied to evaluate the effective diffusion coefficient of an underdamped Brownian particle near the overdamped limit (see details in SI Note 6), for which we once again recover the results obtained via rigorous multiscale analysis^{53,54}. The same method also yields the first-order correction to the effective diffusion coefficient for a particle carrying a dipole moment which interacts with a random or periodic electric field when the dipole dynamics is assumed to be much faster than the particle diffusion⁵⁸.

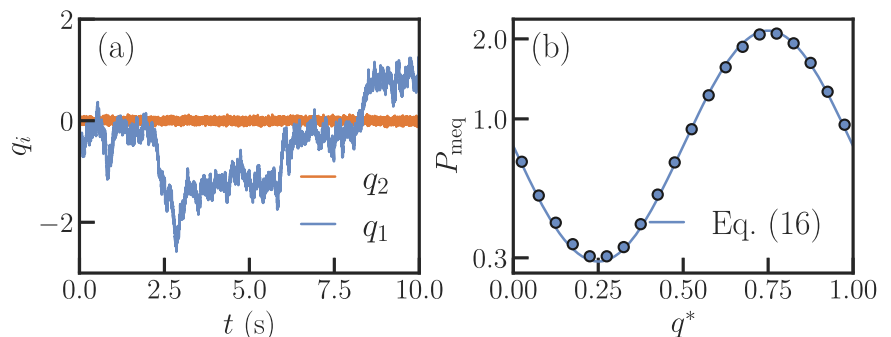
Comparison to numerical simulations

In order to verify the above predictions, Eqs. (3) and (4) are numerically solved for a sinusoidal trapping potential $\phi(q_1)$ (see details in Methods). Typical raw trajectories are shown in Fig. 2a). We observe that the elastic mode $q_2(t)$ fluctuates around its equilibrium value ($q_2 = 0$), within the harmonic potential it is subject to, whereas the position $q_1(t)$ of the test colloid fluctuates at short times in a minimum of the sinusoidal potential $\phi(q_1)$ before randomly jumping from one minimum to the other by passing the potential barrier. From such trajectories, we can first compute the late-time PDF in positions. As an example, Fig. 2b) shows the equilibrium marginal PDF of the periodized variable q^* , defined by $q_1 = q^* \bmod(\lambda)$, which satisfies

$$P_{\text{meq}}(q^*) = \frac{\exp[-\beta\phi(q^*)]}{\int_0^\lambda dq \exp[-\beta\phi(q)]}. \quad (16)$$

Fig. 2 | Trajectories and equilibrium distribution.

a Typical trajectories $q_i(t)$ of the test particle ($i = 1$) and the elastic mode ($i = 2$), simulated by numerically solving Eqs. (3) and (4), for mobility matrix elements $\mu_{11} = 1.0$, $\mu_{12} = 0.5$, $\mu_{22} = 0.5$, with $\Delta U = 1.0$, $\lambda = 1.0$, $\kappa = 1.0 \times 10^{-3}$, time increment $\Delta t = 1.0 \times 10^{-4}$, and $k_B T = 1.0$. Simulations are initialized from equilibrium. **b** Late-time marginal PDF P_{meq} for the periodized position q^* , given by $q_1 = q^* \bmod(\lambda)$, of the test particle, obtained by binning 10^2 trajectories of 10^7 points. The symbols are the results of the numerical simulations. The solid blue line corresponds to the Boltzmann distribution given in Eq. (16).



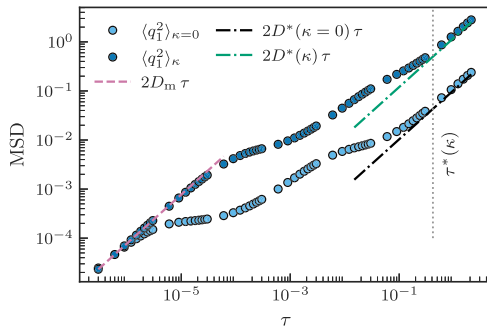


Fig. 3 | Effect of softness on the mean-squared displacement. Mean-squared displacements $\langle q_1^2 \rangle_\kappa$ and $\langle q_1^2 \rangle_0$ of the test particle as functions of the time increment τ , for $\kappa = 0.1$ (dark blue disks) and $\kappa = 0.0$ (light blue disks) respectively, for damping matrix elements $\gamma_{11} = 1.309$, $g_{12} = 1.296$, $\gamma_{22} = 1.309$, with $\Delta U = 2.7$, $\lambda = 0.5$, time increment $\Delta t = 3.0 \times 10^{-7}$, and $k_B T = 1.0$. The symbols are the results of the numerical simulations. The dash-dotted black line indicates $2D^*(0)\tau$, where $D^*(0)$ is the classical Lifson-Jackson late-time diffusion coefficient⁵⁷. The dash-dotted green line indicates $2D^*(\kappa)\tau$, where $D^*(\kappa)$ is the late-time diffusion coefficient given by Eq. (6). The dashed pink line indicates $2D_m\tau$, where $D_m = k_B T \mu_{11}$ is the bare diffusion coefficient. The characteristic time $\tau^*(\kappa) = \lambda^2/D^*(\kappa)$ is indicated with the light-gray vertical dotted line.

In order to probe the dynamical effects of the hydrodynamic coupling of the trapped test particle to the elastic mode, we now turn to the analysis of the mean-squared displacement (MSD), defined as

$$\langle q_i^2 \rangle(\tau) = \langle [q_i(t + \tau) - q_i(t)]^2 \rangle, \quad (17)$$

where $\langle \cdot \rangle$ denotes the ensemble average, and τ is a time increment. As shown in Fig. 3, the MSD of the test colloidal particle exhibits two distinct diffusive regimes separated by a crossover region.

We compute from the MSD the short-time diffusive regime, characterized by the bare diffusion coefficient D_m , and the late-time diffusive regime, which is characterized by the late-time diffusion coefficient D^* , and is reached when the particle has diffused over many periods of the trapping potential, i.e., when its MSD is large compared to λ^2 . This sets a time scale $\tau^*(\kappa) = \lambda^2/D^*(\kappa)$, which must be exceeded to see the late-time diffusive regime. We extract the relative excess in late-time diffusion coefficient, i.e., $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$, as a function of ϵ , from log-log fits of the MSD, excluding: data below $5\tau^*$, which belongs to a different dynamical regime, as well as from the last decade, which lacks statistics for proper data processing. The results are shown in Fig. 4 for a given set of parameters. We observe that the numerical-simulation data agree well with the small- ϵ and large- ϵ asymptotic predictions of Eqs. (6) and (11), respectively. Furthermore, the Kubo formula can be used to evaluate D^* for any value of ϵ (see details in SI Note 5, and Eq. (S65)). We see that the numerical-simulation data agree at all ϵ values with the Kubo formula, and that the latter interpolates well the two asymptotic predictions.

Conclusions

We theoretically and numerically studied the Brownian motion of colloid-like particles near elastic interfaces using minimal ingredients. We showed that a Brownian particle trapped in a periodic potential sees its late-time diffusivity dressed and significantly enhanced by hydrodynamic coupling to a hidden elastic mode. Despite being minimalist, our description reveals generic, original and large effects, relevant for the transport of colloidal particles near complex boundaries. In particular, our results suggest the intriguing possibility of understanding, and even tuning, the diffusivity of colloids near deformable and patchy interfaces—with key implications for biophysics and nanofluidics.

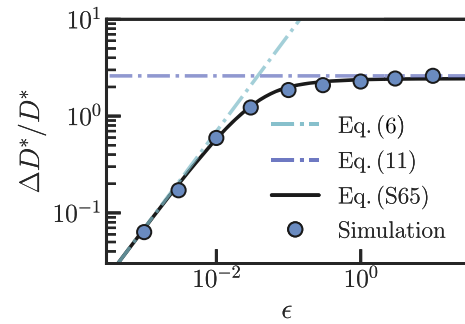


Fig. 4 | Effect of softness on diffusivity. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient D^* of the test particle as a function of the dimensionless compliance ϵ , for $\mu_{11} = 1.0$, $\mu_{12} = 0.85$, $\mu_{22} = 1.0$, $\Delta U = 3.0$, $\lambda = 1.0$, and $k_B T = 1.0$. The symbols are the results of the numerical simulations. The dash-dotted sky blue line corresponds to the small- ϵ prediction of Eq. (6). The dash-dotted navy blue line corresponds to the large- ϵ prediction of Eq. (11). The solid black line corresponds to the Kubo formula of Eq. (S65) (see details in SI Note 5).

Methods

Equations (3) and (4) are numerically solved for a sinusoidal trapping potential $\phi(q_1)$, using an Euler scheme, through a compiled *Cython*⁵⁹ code. The discretized Langevin equations read

$$q_1(t + \Delta t) - q_1(t) = -\mu_{11}\phi'(q_1(t))\Delta t - \frac{\mu_{12}}{\kappa}q_2(t)\Delta t + \xi_1\sqrt{2k_B T\Delta t}, \quad (18)$$

$$q_2(t + \Delta t) - q_2(t) = -\frac{\mu_{22}}{\kappa}q_2(t)\Delta t - \mu_{12}\phi'(q_1(t))\Delta t + \xi_2\sqrt{2k_B T\Delta t}, \quad (19)$$

where ξ_i represent zero-mean Gaussian random variables, characterized by the correlator

$$\langle \xi_i \xi_j \rangle = \mu_{ij}. \quad (20)$$

We carefully choose the time step Δt for numerical stability and accuracy, satisfying $\Delta t \leq 10^{-2}\tau_2$. The mobility tensor μ_{ij} must be positive-definite. To enforce this, we write noises ξ_i as $\xi_i = m_{ij}\eta_j$, with η_j zero-mean, independent Gaussian random variables satisfying $\langle \eta_i \eta_j \rangle = \delta_{ij}$, with $\delta_{ij} = 1$ for $i = j$ and 0 otherwise. This yields a decomposition of the mobility tensor $\mu_{ij} = m_{ik}m_{kj}$. The matrix elements m_{ij} ensure that noise correlations are respected. For computational efficiency, the noise array is pre-generated using the `numpy.random.normal` function. The simulation recursively updates the system state at each discrete time step. Initial positions of both modes are sampled from Eq. (2) to ensure thermalization. Trajectories are computed iteratively, and their corresponding displacement PDFs are binned. Convergence is evaluated with the relative distance between iterations of the stacked displacement PDFs, with a criterion of 1% relative distance between the stacked histogram and new iterations, for more than three consecutive iterations.

Data availability

All data are available upon reasonable request to the corresponding authors. A Supplementary Information (SI) document provides details on: the analytical derivation, and numerical estimations.

Code availability

The simulation scripts are open-source and available on the GitHub repository https://github.com/EMetBrown-Lab/Lacherez2024_Codes. The repository provides a *Jupyter* notebook for single-trajectory simulations. For large-scale parameter sweeps, the `set up.py` script enables optimized

pre-compilation using *Cython*, and the `automate_simulation.py` script supports systematic exploration of the parameter space. These tools are designed for efficient single-node execution and ensure reproducibility of all results presented in this study. We invite the reader to consult the README.md file. Additional treatment codes are available upon reasonable request to the authors.

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Author contributions

J.L. conducted numerical simulations, data analysis and drafted the manuscript. M.L. helped with the optimization of the numerical simulations and commented on the manuscript. Y.A. conceived the project, numerically computed the Kubo formula and commented on the manuscript. D.S.D. conceived the project, developed the analytical model and edited the manuscript. T.S. conceived the project, led the project, and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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